Class 02 activity

Inspect an actuator and compare exact models

21 October 2026 | Team: _____

Each student inspects an unpowered XC330 and returns it. Teams use the official model documents or the lesson's attributed extract. Keep power disconnected; do not disassemble the unit or force its output.

1. Inspect and identify

Exact model: _____

Label the output, fixed mounting and two bus ports in a small sketch.

Which part moves relative to the housing? _____

Available feedback, checked in the documentation: _____

_____

2. Compare XC330-M288-T and XL330-M288-T

Include units and conditions for numerical values. Identify the source.

Comparison	Your evidence
One shared property	_____
Construction or mass	_____
Torque or speed	_____
Source / section	_____

An intended use and the property that matters: _____

_____

Can stall torque and no-load speed occur together? Why? _____

_____

Does encoder resolution establish accuracy? Explain. _____

_____

3. Record and return

All units returned: _____ Team comparison and labels recorded: _____

No powered student operation, software installation or telemetry calculation is required. Optional follow-up: locate three research examples using XC330/XL330; record the exact model and role, or mark the exact model unverified.